DAY 33 | PRE-READ

From Prototype to Production Service

A light read before Day 33. About 15 minutes, no code required.

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Why read this?

A prototype that works for one person in a notebook is very different from a service that answers thousands of users safely. Tomorrow you learn how AI services are deployed, configured, scaled and updated without breaking things.

1. Rent, lease or own

Option	Everyday comparison	Trade-off
Use a model provider's API	Taking a taxi	Easy and powerful; you follow their rules and pay per trip
Host it in your company's cloud	Leasing a car	More control and privacy; some setup
Run an open model yourself	Owning a car	Full control; you handle fuel, servicing and repairs

2. Practice kitchen and real kitchen

Restaurants test new dishes in a practice kitchen before serving customers. Software teams use separate **environments**: development for building, testing for checking, production for real users. Passwords and keys, called **secrets**, are kept in a safe, never written into the recipe.

3. Serving many people at once

A good waiter does not stand at the kitchen door waiting for one dish; they take other orders meanwhile. AI services work the same way, handling many requests while waiting for the model. They also **stream** answers word by word, so users see progress immediately.

4. Changing safely

Before switching every customer to a new menu, a restaurant might offer it at a few tables first. AI teams do this too, called a **canary release**: a small share of users get the new version while the team watches the numbers.

Key idea: release changes gradually and make it easy to switch back.

5. Think about these before class

- Would your organisation allow its data to be sent to an external AI provider? What would they need to know?
- Which settings would differ between a test and a live version of your capstone?
- Why does seeing an answer appear word by word feel faster, even if the total time is the same?

Nagareru (流れる): to flow